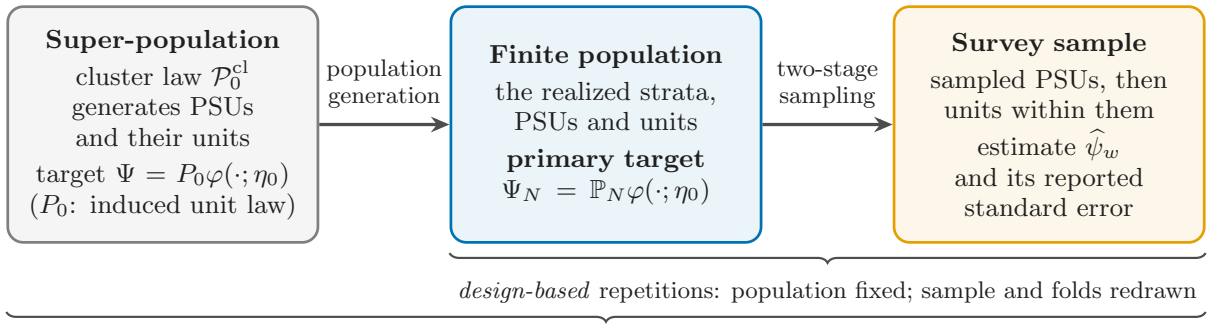

Super-population
cluster law $\mathcal{P}_0^{\mathrm{cl}}$
generates PSUs
and their units
target $\Psi = P_0\varphi(\cdot;\eta_0)$
(P_0: induced unit law)
population
generation
Finite population
the realized strata,
PSUs and units
primary target
$\Psi_N = \mathbb{P}_N\varphi(\cdot;\eta_0)$
two-stage
sampling
Survey sample
sampled PSUs, then
units within them
estimate $\widehat{\psi}_w$
and its reported
standard error
design-based repetitions: population fixed; sample and folds redrawn
joint model–design repetitions: population regenerated, then sampled